

Evaluating Cybersecurity Vulnerabilities in SMPTE ST 2110 Media Networks

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Abstract –This paper presents a threat-lab environment for empirically evaluating cybersecurity risks in SMPTE ST 2110 and AMWA NMOS, based IP media workflows. A segmented testbed, comprising dedicated media, control, and management VLANs; a router VM; and Raspberry Pi nodes acting as sender, receivers, adversary, and monitors, enables controlled execution of representative attack scenarios. These include Layer-2 manipulation (ARP spoofing, MAC flooding), multicast/IGMP disruption (rogue querier, join/report floods), RTP spoofing and payload replacement, and NMOS control-plane interference. Synchronized multi-vantage PCAPs and logs provide temporally aligned measurements of RTP loss, jitter, skew, SSRC behavior, IGMP state transitions, and NMOS heartbeat stability. Across experiments, results consistently show that multicast media playout can appear visually stable even as control-plane signaling and timing degrade. Attacks such as ARP poisoning, IGMP floods, and NMOS HTTP saturation produced significant jitter excursions, registry instability, and forwarding anomalies, yet often with minimal immediate impact on perceived video quality. These findings highlight a critical gap between operational perception and underlying network health. The paper concludes by mapping observed failure modes to practical mitigations and emphasizing the need for robust telemetry and defense-in-depth designs as ST 2110 facilities scale.

Introduction

The broadcast industry is undergoing a significant transition from traditional SDI infrastructures to IP-based media workflows, most prominently those defined by the SMPTE ST 2110 suite of standards. This shift is driven by the need for greater scalability, flexible routing, and deeper alignment between broadcast operations and modern IT practices. Moving media transport onto IP fundamentally changes how signals are routed, monitored, and synchronized, replacing dedicated hardware paths with software-defined and network-centric architectures. At the same time, the transition removes many of the physical and electrical isolation properties inherent to SDI, introducing cybersecurity considerations that were not previously relevant in closed, point-to-point environments.

A key feature of ST 2110 systems is the decomposition of video, audio, and ancillary data into independent IP streams, coordinated through a common timing reference such as IEEE 1588 Precision Time Protocol. While the ST 2110 standards define how these essences are transported across an IP network, they do not specify mechanisms for discovery or connection management. These capabilities are provided by the AMWA Networked Media Open Specifications (NMOS), which introduce REST-based APIs for resource registration, device

discovery, and sender/receiver binding. Together, ST 2110 and NMOS enable flexible, software-driven orchestration of live media systems, but in doing so, they expand the attack surface beyond what has historically existed in baseband environments.

As IP-based media networks incorporate software-defined routing, multicast-driven distribution, and open control interfaces, vulnerabilities emerge across multiple layers of the media stack. These include weaknesses in link-layer behavior, multicast and IGMP signaling, RTP media transport, and NMOS control-plane operations. Such vulnerabilities have the potential to disrupt real-time media flows, corrupt content, or interfere with production-critical coordination.

Previous studies examining security within IP-based media infrastructures highlight similar concerns about the exposure created by the transition from SDI to networked workflows. Industry analyses from the European Broadcasting Union [1] and AMWA [2] emphasize that ST 2110 and NMOS were designed primarily for interoperability and timing precision rather than adversarial resilience, leaving essential control-plane functions, such as IGMP membership management, ARP resolution, and NMOS routing, without authentication or integrity protections. Academic work on multicast reliability supports this vulnerability profile: Jalal and Li [3] demonstrated that IGMP control logic can be disrupted through forged joins and rogue querier elections, while Al-Shuwaili et al. [4] showed that IGMP membership suppression can destabilize real-time multicast delivery. Research into Ethernet-layer attacks further underscores these risks. Arslan [5] and Bull [6] documented how ARP spoofing, MAC-table poisoning, and VLAN hopping can be used to misdirect or intercept traffic on switched networks, with direct implications for uncompressed ST 2110 media flows. More recent studies have explored weaknesses in API-driven broadcast-control systems: the NMOS security guidance [7] and Thiagarajan et al. [8] highlight that NMOS's REST interfaces, when deployed without HTTPS or authorization controls, can permit unauthorized connection changes or resource manipulation. While these efforts independently characterize vulnerabilities across Layer-2, multicast, and application-control layers, few provide an integrated, empirical evaluation of how such attacks manifest within a functioning ST 2110 environment. This gap motivates the experimental analysis presented in this work.

Evaluating these risks requires controlled experimentation that reflects operational ST 2110 behavior. This work addresses that need by defining a threat model for IP-based media systems and analyzing representative attacks within a purpose-built ST 2110 simulated testbed. The following sections outline the threat model, experimental methodology, and observed impacts of adversarial activity on media transport and control-plane stability.

Threat Lab

To structure the analysis that follows, a clear threat model is required to describe the assumptions, adversarial capabilities, and attack surfaces relevant to IP-based media systems. The model considers an adversary capable of injecting, modifying, or redirecting traffic within the media network, reflecting realistic internal threat scenarios observed in operational facilities. The threat lab developed for this study evaluates how such an adversary can exploit vulnerabilities across Layer-2 switching behavior, multicast and IGMP control signaling, and application-layer protocols such as RTP and NMOS. These categories collectively define the

scope of the attacks examined in this work and form the basis for the experimental scenarios evaluated in the subsequent sections.

The relevance of these vulnerabilities is underscored by industry security guidance. The European Broadcasting Union's cybersecurity recommendations for media workflows highlight that modern IP-production systems inherently depend on unauthenticated Layer-2 mechanisms, multicast control signaling, and unencrypted control-plane APIs, making them especially susceptible to internal compromise and lateral movement. The EBU stresses that insufficient network segmentation, poor isolation of media and control domains, and the widespread use of trust-based device discovery create conditions in which a single compromised endpoint can exert disproportionate influence over media transport and orchestration workflows [14]. These industry findings directly motivate the adversarial assumptions adopted in this work.

In operational broadcast environments, these attacks typically originate from a compromised or unauthorized device that has gained access to the media network. Such access may result from infected engineering workstations, misconfigured NMOS nodes, unsecured switch ports, or insider threats in facilities where production networks share physical or logical boundaries with general-purpose IT infrastructure. Critically, both academic and industry security analyses emphasize that the most impactful attacks often come from inside the trusted Layer-2 domain. Cisco's SAFE security architecture notes that organizations routinely overlook "insider abuse, misuse, or misconfigured internal devices" as major sources of compromise, even though these represent some of the most common and damaging threat vectors on enterprise networks [15]. Likewise, research into ARP spoofing demonstrates that Layer-2 protocols in Ethernet LANs were designed without authentication, allowing a malicious host on the same subnet to trivially forge ARP replies, poison caches, and perform man-in-the-middle interception of high-bandwidth traffic, attacks that standard firewalls and antivirus tools cannot detect [16]. Because ST 2110 media workflows frequently operate within flat or minimally segmented Layer-2 domains, an adversary with media-VLAN visibility can exploit unauthenticated protocols such as ARP, IGMP, RTP, and NMOS to manipulate forwarding state, inject multicast traffic, or disrupt control-plane behavior. The threat model used in this study reflects this assumption by positioning the adversarial node within the media subnet, representing a realistic scenario in which a compromised host operates with full visibility into multicast distribution and control signaling.

Layer-2 [19] vulnerabilities arise from the absence of authentication, integrity checks, and source validation in fundamental Ethernet mechanisms, a limitation widely documented in SANS Institute security guidance, which notes that traditional LAN protocols offer no inherent protection against spoofing or traffic manipulation [13]. As a result, Address Resolution Protocol (ARP) messages can be trivially forged, allowing an attacker to impersonate gateways or endpoints and redirect high-bandwidth media flows without detection. The same lack of validation enables adversaries to poison switch forwarding tables by injecting fraudulent MAC entries or flooding the switch with bogus source addresses, forcing it into fallback broadcast behavior. Furthermore, VLAN boundaries can be undermined through mis-tagging or double-tagging techniques, exposing media and control traffic that would otherwise remain

isolated. Taken together, these weaknesses form the basis of several well-known Layer-2 attacks, including ARP spoofing, MAC table exhaustion, and VLAN mis-tagging, that can destabilize multicast forwarding, disrupt routing behavior, and degrade the integrity of ST 2110 media flows.

Beyond Layer-2 behavior, multicast distribution introduces its own security risks, as ST 2110 workflows rely heavily on IGMP signaling to manage group membership and forwarding state. Industry security guidance on secure development practices emphasizes that unauthenticated control-plane protocols constitute a persistent attack vector in modern networked environments [17], underscoring the susceptibility of IGMP to spoofed membership reports, rogue querier elections, and other forms of manipulation. Because IGMP membership reports directly influence switch forwarding tables, a malicious host can disrupt or suppress multicast delivery by issuing forged joins, leaves, or reports that destabilize group membership state. In addition to falsified membership reports, an attacker may exploit the IGMP querier election process to assume control of multicast group management, resulting in inconsistent or erroneous forwarding state. Furthermore, excessive or malformed IGMP messages can overload switch control planes, degrading membership processing and jeopardizing real-time media delivery. Such attacks can result in unpredictable forwarding behavior, loss of media flows, or excessive control-plane load, all of which undermine the stability of ST 2110 multicast transport.

Application-layer components in ST 2110 systems, particularly RTP transport and NMOS control-plane APIs, are vulnerable to manipulation because their packet formats and REST interfaces *lack inherent mechanisms* for verifying source authenticity or preventing unauthorized modification, a limitation highlighted in the NMOS HTTPS security guidance, which notes that HTTP messages provide no native authentication, integrity, or confidentiality protections [18]. In practice, this exposes both media and control workflows: attackers can spoof or inject RTP packets to distort media content, and they can exploit unauthenticated NMOS APIs to alter routing state or introduce falsified resources. The resulting loss of trust in both media payloads and control transactions presents a high-impact failure mode, particularly in environments that rely on deterministic routing and real-time delivery guarantees. Moreover, the lack of embedded security or verification features in these protocols limits the ability of monitoring systems to distinguish between legitimate behavior and malicious interference. Taken together, these weaknesses highlight the need to rigorously evaluate application-layer behavior under adversarial conditions, as even limited manipulation can directly compromise media integrity and system control.

Experimental Methodology

To experimentally evaluate the vulnerabilities identified in the threat model, a controlled simulated ST 2110 test bed was constructed to replicate the essential media-transport, multicast, and control-plane behaviors of a modern IP production environment. Media traffic was generated as MPEG-TS over RTP to provide a practical stand-in for uncompressed ST 2110 flows while preserving the same IP multicast, RTP timing, and control-plane characteristics relevant to security analysis. The arrangement allows for deterministic routing of media streams while providing visibility into multicast behavior, control-plane signaling, and adversarial traffic

injection across distinct network segments. The nodes are assigned distinct functional roles within the topology, with dedicated devices handling media transmission, reception, adversarial activity, and system monitoring to ensure clear separation of responsibilities during each experiment. By organizing these nodes across separate media, control, and management segments, the testbed preserves the logical separation found in production environments and facilitates targeted analysis of attack propagation. Together, these design choices ensure that the resulting observations reflect both the operational characteristics of ST 2110 systems and the specific effects of the adversarial conditions under study.

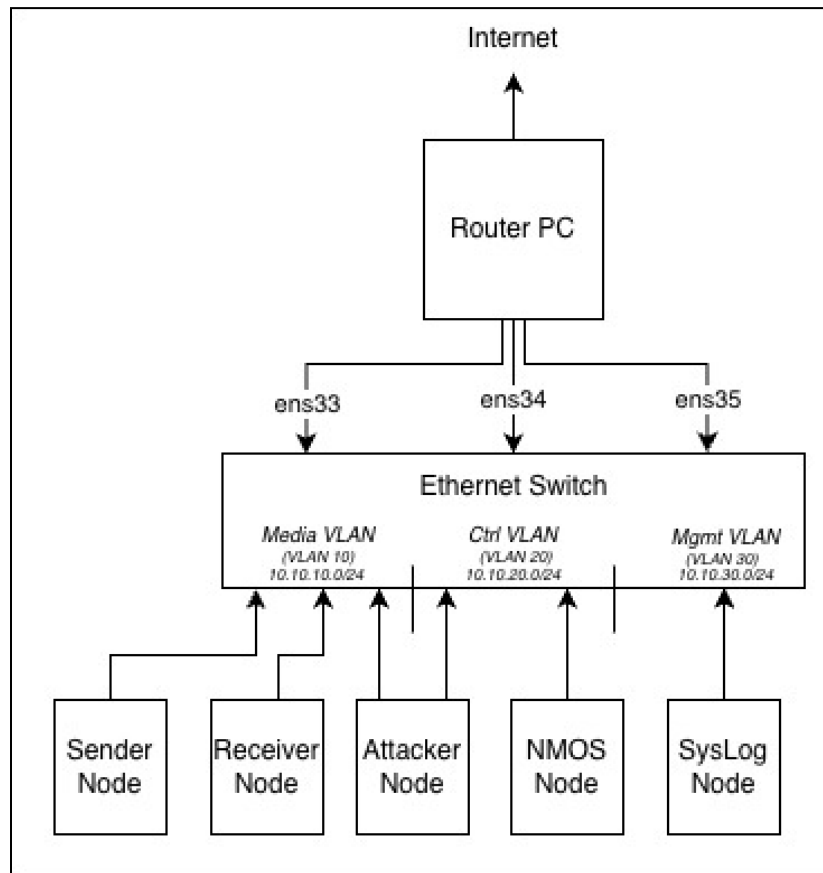


FIGURE 1: High-Level Model of Threat Lab

To accurately reflect operational ST 2110 deployments, the testbed employs a segmented network configuration that isolates media flows, control-plane signaling, and management operations across distinct VLAN domains. By separating these traffic classes, the testbed reproduces the logical isolation commonly used in production facilities to maintain predictable media performance and limit the blast radius of control-plane or management-plane faults. A router virtual machine provides controlled inter-VLAN forwarding, enabling media, control, and management domains to interact only through explicitly defined pathways. Within the media VLAN, multicast distribution is governed by IGMP snooping and querier functionality, allowing the testbed to model the forwarding behavior typical of ST 2110 environments. While the testbed

does not incorporate PTP timing or full-facility orchestration, it captures the essential forwarding, multicast, and control-plane behaviors relevant to the attack scenarios under study. This configuration enables predictable multicast group formation and ensures that sender and receiver behavior can be evaluated under controlled membership and forwarding conditions.

The testbed consists of discrete sender, receiver, adversary, and monitoring nodes, each implemented on dedicated Raspberry Pi or virtualized hosts with hardware and software configurations tailored to their specific functional roles. The sender, receiver, and adversary roles were deployed on Raspberry Pi 4B devices running Raspberry Pi OS Lite, providing consistent ARM-based performance and a uniform network interface for media transport and attack execution. The router and monitoring functions were hosted on a Linux Ubuntu Server virtual machine running on VMWare Workstation Pro, which offered multi-interface support, increased processing capacity for logging and analysis, and a controlled environment for inter-VLAN routing. The media sender used an FFmpeg-based RTP over UDP pipeline, while receivers employed lightweight decoding and logging tools, enabling consistent validation of playout behavior and packet integrity under varying network conditions. Together, these hardware and software choices provided a stable and reproducible platform for evaluating each attack scenario.

To ensure consistent, repeatable observation of media and control-plane behavior across all experiments, the testbed employed a synchronized, multi-node capture and logging framework that aligned packet traces and system events in time. Capture synchronization was achieved through a UTC-epoch scheduling mechanism, in which all nodes waited for a predetermined start time before initiating their packet captures and logging routines. To maintain a clean baseline at the beginning of each run, sender traffic was initially idle and then triggered to start five seconds after the capture window opened. Packet captures were collected simultaneously on the sender, receivers, adversary node, router VM, and monitoring VM, providing a comprehensive view of media-path behavior and control-plane activity across the entire topology.

Traffic at each vantage point was recorded using *tshark* with consistent capture parameters to ensure uniform visibility across hardware. In addition to packet traces, each node exported system-level logs, including *dmesg*, *journalctl* output, interface statistics, and NMOS registry snapshots, to capture control-plane events and device state changes associated with each experiment. This unified capture and logging approach provided tightly correlated temporal data, enabling reliable comparison between baseline and adversarial conditions and ensuring that observed deviations could be attributed directly to the attack under study. Each experiment followed a consistent procedure in which the testbed was returned to a known baseline state, synchronized captures were initiated, and the corresponding adversarial behavior was introduced only after the media stream had stabilized. This repeatable execution pattern ensured that observed deviations could be attributed solely to the injected attack rather than background variability. The combination of controlled topology, synchronized capture, and consistent execution provides a reliable basis for the comparative analysis presented in the following section.

Expected Attack Behavior

Layer-2 Attack Behavior

Layer-2 attacks are expected to immediately affect reachability and forwarding within the media VLAN because ARP, MAC learning, and VLAN tagging provide no authentication. An adversary on the subnet can redirect or corrupt traffic by forging ARP replies, exhausting MAC tables, or injecting mis-tagged frames. ARP spoofing should cause receivers to lose access to the legitimate sender or route RTP flows through the attacker, producing packet loss or playout failure. MAC exhaustion is expected to trigger broadcast flooding, increasing congestion and degrading real-time delivery. VLAN mis-tagging may also expose traffic between domains. These effects should appear in captures as abnormal MAC movements, duplicate unicast traffic, and elevated packet loss.

Expected IGMP/Multicast Attack Behavior

Multicast attacks are expected to destabilize ST 2110 distribution by exploiting unauthenticated IGMP signaling. Because joins and leaves directly determine forwarding state, a malicious host can disrupt delivery by issuing forged or excessive reports. Join floods may overload the switch's control plane, while spoofed leaves can prematurely remove receivers from multicast groups, causing immediate loss of media. A rogue querier is expected to disturb refresh cycles and create inconsistent group membership. In captures, these effects should appear as irregular IGMP activity, fluctuating memberships, unexpected multicast flooding or suppression, and interruptions in receiver continuity.

Expected RTP and NMOS Attack Behavior

At the application layer, RTP and NMOS attacks are expected to produce visible changes in stream integrity and routing behavior. Since RTP lacks source authentication, injected packets may overwrite or interleave with legitimate media, causing corrupted frames, sequence errors, jitter spikes, or SSRC conflicts. Malformed packets may also disrupt receiver timing. NMOS attacks, enabled by unauthenticated HTTP control APIs, are expected to cause unauthorized routing changes, falsified resource records, or unexpected connection modifications. These behaviors should manifest as anomalous REST traffic, unexpected NMOS JSON updates, and transport disruptions not attributable to lower-layer faults.

Scenarios

This section outlines the adversarial scenarios executed in the ST 2110 testbed to evaluate how specific Layer-2, multicast, and application-layer attacks manifest under controlled experimental conditions. Each scenario targets a distinct vulnerability identified in the threat model, enabling its effects on media transport, control-plane signaling, and system behavior to be observed in isolation. By executing these attacks within a synchronized and instrumented environment, the study provides a controlled basis for comparing baseline operation against adversarial conditions and for quantifying the specific failure modes introduced by each threat.

#	Category	Experiment Name	Description
1	Baseline	Baseline 2110 flow	Clean capture of normal traffic
2	L2 Attack	ARP spoof / gateway hijack	Pi4 impersonates router, observe disruptions
3	L2 Attack	MAC table flood	Flood switch with fake MACs, observe packet loss
4	Multicast/IGMP	Normal IGMP behavior	Verify join/leave handling, MFDB, and expected multicast paths.
5	Multicast/IGMP	Rogue IGMP querier takeover	Pi4 becomes querier; check membership instability
6	Multicast/IGMP	Join/leave/report floods	Flood IGMP joins/reports; track switch CPU, counters, jitter, loss.
7	RTP Spoofing	Basic RTP injection	Inject fake SSRC into the same group/port
8	RTP Spoofing	Payload replacement	Inject alternate video payload
9	NMOS Spoofing	NMOS registry manipulation	Insert fake nodes/flows, demonstrate mis-routing or false devices.

TABLE 1: Summary of Experimental Scenarios

Layer 2 Attack Scenarios

The Layer-2 attack scenarios evaluate how unauthenticated Ethernet mechanisms like ARP resolution and MAC learning, can be exploited by a malicious host to disrupt or redirect ST 2110 media flows. These scenarios focus on attacks, such as ARP spoofing, MAC-table exhaustion, and VLAN mis-tagging, each selected to isolate a distinct failure mode and reveal its impact on real-time media transport. By exercising these fundamental link-layer weaknesses, the experiments demonstrate how even simple manipulations at Layer-2 can produce immediate and observable disruptions across the entire media path.

Multicast/IGMP Attack Scenarios

The multicast and IGMP attack scenarios examine how unauthenticated group-management signaling can be manipulated to destabilize ST 2110 multicast distribution and disrupt the delivery of real-time media flows. These scenarios involve forged membership reports, spoofed leave messages, join floods, and rogue-querier behavior, each designed to expose how multicast forwarding state can be corrupted or suppressed by a malicious host. Together, these tests illustrate how small perturbations to IGMP control traffic can trigger disproportionate disruptions in multicast routing, resulting in loss, flooding, or inconsistent delivery across receivers.

RTP/NMOS Spoof Attack Scenarios

The RTP and NMOS attack scenarios evaluate how unauthenticated media transport and control-plane APIs can be manipulated to alter, corrupt, or redirect ST 2110 streams at the application layer. These scenarios include RTP packet injection, sequence-number manipulation, SSRC spoofing, and unauthorized NMOS routing or resource modifications, each

chosen to reveal how application-layer tampering can affect stream integrity and control-plane state. By targeting the protocols responsible for media integrity and connection management, these tests demonstrate how application-layer attacks can produce content corruption, unintended routing changes, or loss of orchestration awareness even when lower network layers remain stable.

Criteria and Metrics

To assess the impact of the adversarial scenarios executed in the ST 2110 testbed, a consistent set of metrics was collected across all experiments. These measurements capture how Layer-2 manipulation, multicast disruption, RTP interference, and NMOS control-plane spoofing affect media transport stream and behavior of the systems. Each metric was selected to reflect observable failure modes in professional media networks and to allow direct comparison between baseline operation and adversarial conditions.

Transport-layer metrics: RTP packet loss, jitter, sequence errors, SSRC collisions, and payload integrity, were used to quantify the degradation of media delivery under attack. Receiver-side indicators such as frame freezes, decode errors, and continuity counter jumps provided evidence of perceptible impact on playout quality. Multicast behavior was evaluated through IGMP join/leave timing, MFDB stability, and the presence of unintended flooding, enabling analysis of how attacks affect group membership and distribution paths.

Network level metrics: interface counters (drops, overruns, errors) and switch CPU utilization capture stress induced by MAC flooding or IGMP storms. Control-plane integrity was assessed through NMOS heartbeat stability, registry consistency, and the detection of unauthorized devices or manipulated flow entries. For each experiment, the attack was considered successful when it produced measurable deviations from baseline values, induced instability in multicast or routing behavior, corrupted RTP flows, or caused visible disruption at the receiver.

This evaluation framework provides a repeatable and objective basis for comparing the severity of each threat and its corresponding effect on 2110 media delivery, forming the foundation for the analysis presented in the subsequent section.

Scenario Experiment Results

Baseline Scenario

During this baseline scenario, Pi1 transmitted a continuous MPEG-TS RTP stream to Pi2 under normal operating conditions, with all nodes using the default gateway at 10.10.10.1 and no adversarial activity present. Across all trials, the sender and receiver consistently reported matching SSRC values (0xf3001e1f, 0x891af0fe, 0xf3001e1f), confirming a stable, single-source media flow throughout the measurement period. Jitter remained low and consistent, with maximum values of approximately 12.12 ms and mean jitter near 0.26 ms on

both sender and receivers across the 3 trials, indicating an uncongested and deterministic transport path. No packet loss or sequence errors were observed at either endpoint, and RTP packet counts matched expected values, demonstrating clean multicast delivery across the router VM and switch. Clock drift remained minimal (≈ -0.66 ms), consistent with free-running Pis operating without PTP, and did not impact media stability. NMOS heartbeats and registration traffic were uninterrupted for the full duration, confirming steady Layer-3 control-plane behavior. Overall, the baseline conditions produced a fully stable 2110 workflow with no observable anomalies, forming a reliable reference point for evaluating the impact of subsequent adversarial scenarios.

ARP Spoofing Scenario

During the ARP spoof/gateway hijack test, Pi4 generated unsolicited ARP replies that impersonated the router's IP address (10.10.10.1), overwriting the legitimate gateway MAC entry on the sender and receiver. As shown in Fig. 2, the first spoofed ARP reply appeared at approximately $T \approx 23$ s, with Pi4 repeatedly asserting "10.10.10.1 is-at dc:a6:32:71:2b:f7," indicating a successful poisoning of the ARP caches. Immediately after the spoof began, ICMP reachability to both Pi1 and Pi2 failed, and NMOS heartbeat traffic stopped, confirming loss of Layer-3 control-plane connectivity. RTP analysis on the receiver showed a corresponding freeze and packet loss burst during the attack window; however, the router VM continued to receive uninterrupted multicast video because ST 2110 media distribution is forwarded by the switch at Layer-2 and does not require ARP-based gateway resolution. These results demonstrate that a simple ARP poisoning event can disrupt NMOS and degrade receiver-side RTP performance without immediately interrupting multicast delivery at other network vantage points.

3428	22.011165742	RaspberryPiT_71:2b..	Broadcast	ARP	60	Who has 10.10.10.11? Tell 10.10.10.13
3429	22.011245386	RaspberryPiT_39:b9..	RaspberryPiT_71:2b..	ARP	42	10.10.10.11 is at dc:a6:32:39:b9:35
3439	22.040513377	RaspberryPiT_71:2b..	Broadcast	ARP	60	Who has 10.10.10.12? Tell 10.10.10.13
3665	23.011713009	RaspberryPiT_71:2b..	RaspberryPiT_39:b9..	ARP	60	10.10.10.1 is at dc:a6:32:71:2b:f7
4032	25.012233906	RaspberryPiT_71:2b..	RaspberryPiT_39:b9..	ARP	60	10.10.10.1 is at dc:a6:32:71:2b:f7
4364	27.012433117	RaspberryPiT_71:2b..	RaspberryPiT_39:b9..	ARP	60	10.10.10.1 is at dc:a6:32:71:2b:f7
4378	27.054500672	RaspberryPiT_39:b9..	RaspberryPiT_71:2b..	ARP	42	Who has 10.10.10.13? Tell 10.10.10.11
4379	27.054623428	RaspberryPiT_71:2b..	RaspberryPiT_39:b9..	ARP	60	10.10.10.13 is at dc:a6:32:71:2b:f7

FIGURE 2. ARP Spoofing Traffic Captured During Gateway Hijack Test

Figure 2 - Representative ARP packets captured on Pi1 during the spoofing attack. Normal ARP requests (frames 3428–3439) are followed by unsolicited forged replies in which Pi4 advertises "10.10.10.1 is-at dc:a6:32:71:2b:f7," confirming the gateway hijack.

MAC Table Flooding Scenario

During the MAC table flooding experiment, the switch's forwarding database expanded rapidly from approximately 15 learned MAC entries to over 800 within a single minute, confirming that the attacker successfully executed a sustained CAM-exhaustion event. Despite this aggressive increase in L2 churn, RTP transport metrics remained unexpectedly stable, with both the sender and receiver reporting zero packet loss, no sequence errors, and nearly identical packet counts. Jitter stayed within the microsecond range, with mean jitter around $0.24 \mu\text{s}$ and maximum jitter near $12 \mu\text{s}$, indicating that multicast forwarding continued deterministically even as the switching

fabric was stressed. Maximum skew remained under 400 μ s for both endpoints, showing that timing alignment and inter-arrival spacing were not disrupted under load. Receiver logs contained expected H.264 PPS/SPS warnings from joining mid-GOP, but no decoder failures or playout disruptions attributable to network impairment. Switch statistics reflected elevated MAC-learning activity but did not show the broadcast storms or unknown-unicast surges typically associated with severe CAM exhaustion.

Overall, the results demonstrate that while the MAC flood significantly stressed the switch's L2 state, the multicast delivery path for the 2110-like stream remained intact, highlighting the inherent resilience of IGMP-controlled forwarding even under adversarial conditions.

IGMP Baseline Scenario

The IGMP baseline capture demonstrates correct multicast group management behavior in the ST 2110 testbed. When the receiver (10.10.10.12) subscribed to the media stream, it issued an IGMPv2 Membership Report for group 239.10.10.10 at approximately 0.61 seconds, indicating a clean and timely join event. At 6.12 seconds, the receiver transmitted an IGMP Leave message, followed immediately by a group-specific query from the switch and a new Membership Report, reflecting proper querier behavior and fast pruning-rejoin cycles. The periodic Membership Reports that follow confirm stable group membership with no evidence of churn or unexpected transitions. Complementary PCAP analysis showed zero packet loss, consistent SSRC values, and low mean jitter, indicating that multicast forwarding remained stable throughout the join/leave sequence. Additionally, no UDP traffic leaked to non-member ports, confirming that IGMP snooping prevented multicast flooding and that the switch correctly pruned inactive interfaces. Together, these results validate that the multicast domain behaved deterministically under normal conditions, providing a reliable baseline for comparison against adversarial IGMP manipulation in later experiments.

38	0.615047281	10.10.10.12	239.10.10.10	IGMPv2	46 Membership Report group 239.10.10.10
57	6.122760703	10.10.10.12	224.0.0.2	IGMPv2	46 Leave Group 239.10.10.10
60	6.135110327	10.10.10.12	239.10.10.10	IGMPv2	46 Membership Report group 239.10.10.10
753	10.367039704	10.10.10.12	239.10.10.10	IGMPv2	46 Membership Report group 239.10.10.10

FIGURE 3. IGMPv2 Membership Reports and Leave Messages Observed During Baseline Multicast Join/Leave Behavior

This IGMP trace in Figure 3 shows the receiver issuing a timely Membership Report to join multicast group 239.10.10.10, followed by a clean Leave and immediate re-join triggered by the querier's group-specific query. The sequence confirms correct IGMPv2 operation, stable group membership, and proper leave/prune behavior under baseline conditions.

Rogue IGMP Querier Scenario

The rogue querier experiment produced a clear shift in IGMP control-plane behavior without causing packet loss on the ST 2110 media flow. Before the attack, the Router VM at 10.10.10.1 issued periodic general queries, but once the attacker (Pi4 at 10.10.10.13) began transmitting rapid IGMPv2 General Queries, the receiver's PCAP showed a noticeable increase in Membership Reports and group-specific responses aligned with the attacker's timing. This

resulted in bursts of IGMP activity and visible churn in the group maintenance logic, with receivers repeatedly issuing Membership Reports immediately after Pi4's queries in the 23–36 second range. Despite this control-plane instability, RTP transport remained intact: sender and receiver statistics both reported 0% packet loss, no sequence errors, and consistent SSRC continuity.

The primary observable effect was a rise in short-term jitter, with a maximum jitter of ~12 ms and a corresponding spike in inter-packet delta values around the takeover period, although mean jitter remained low (~0.26 ms). No evidence of MFDB collapse or multicast flooding appeared in the captures, suggesting that the switch maintained forwarding entries even as the querier role shifted. The media stream did not suffer sustained degradation, only minor timing variation, indicating that rogue querier manipulation can disturb IGMP timing but does not necessarily force dataplane failure in a small-scale 2110 network. Overall, the results demonstrate that the IGMP control plane was successfully hijacked by the attacker, but the underlying multicast forwarding path showed strong resilience under these specific conditions.

IGMP Join/Leave/Report Flood Scenario

The IGMP join/leave/report flood produced measurable control-plane stress on the switch, reflected in increased CPU-interface unicast counters and rising octet-level activity during the attack window. Although the Netgear platform does not expose CPU utilization as a percentage, the deltas across snapshots show approximately 1,000 additional CPU-bound packets and over 118 KB of additional control-plane traffic processed during the 52-second flood, confirming that IGMP messages were consistently punted to the switch processor. This control-plane load correlated directly with degradation in RTP transport stability at the receivers. While no packet loss or sequence errors were recorded, both RTP sessions exhibited elevated timing irregularities, including a 12× increase in maximum jitter, ~20× increase in maximum delta inter-arrival times, and cumulative skew drifting as high as 350 ms. These values represent significant deviation from baseline ST 2110 behavior, where timing variation typically remains sub-millisecond. The experiment therefore demonstrates that Layer-2 IGMP control-plane flooding can induce substantial timing instability in otherwise lossless media streams. Even without IGMP snooping enabled, the switch's CPU must still parse, classify, and discard IGMP control packets, and this additional workload is sufficient to disrupt receiver playout timing. Collectively, these results highlight the susceptibility of multicast-dependent media workflows to control-plane disruption, even when endpoint packet delivery remains superficially lossless.

RTP Payload Spoofing Scenario

During the RTP payload-replacement attack, VLC continued rendering the legitimate ST 2110-like stream without visible disruption, even as transport-layer timing became heavily distorted. ffmpeg and Wireshark recorded significant anomalies, multiple unexpected SSRCs, jitter spikes exceeding 20–30 ms, and large negative skew values, indicating that the attacker was successfully injecting poorly synchronized RTP traffic into the multicast group. The attack script fully replicated the original sender's source identity at Layer 2 and Layer 3, spoofing both the MAC address and IP address of the legitimate Pi1 transmitter, ensuring that injected packets were indistinguishable from the authentic flow at the network level. The legitimate sender's

SSRC remained stable throughout the test, with negligible jitter (~ 0.24 ms) and no packet loss, giving it a strong timing advantage over the injected flows. In contrast, the attacker-generated SSRCs exhibited massive frequency drift, incorrect clock rates (~ 31 kHz instead of 90 kHz), and severely unstable timing profiles. This imbalance caused the receiver's jitter buffer to reject most attacker packets, preventing a full “stream flip” despite heavy contamination on the wire. Brief corruption was observed at approximately 1–2 seconds, but the receiver restabilized within ~ 30 seconds as VLC locked back onto the clean, long-running reference stream. Across repeated trials, the qualitative behavior was consistent: the attack impacted transport metrics but never achieved decoder takeover. These results show that a successful RTP replacement attack requires not only SSRC spoofing, but also precise alignment of sequence numbers, timestamps, and clock frequency, even when MAC and IP identity are perfectly forged.

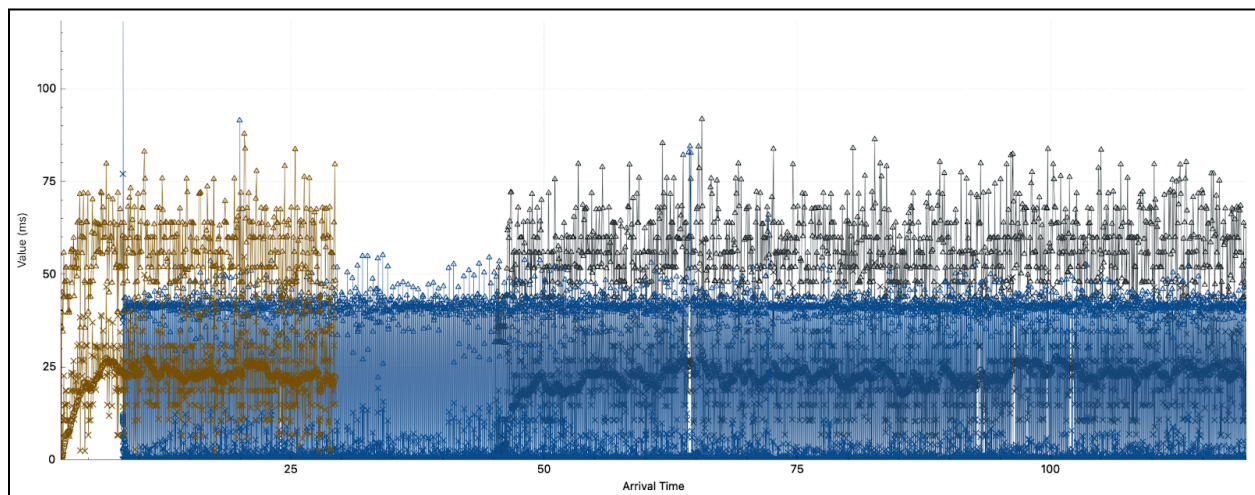


FIGURE 4. RTP Inter-Arrival Time and Jitter Distribution Over Experiment Duration

The arrival-time plot illustrates the severe transport-layer instability introduced during the RTP payload-spoofing experiment. The legitimate sender's packets (blue) form a dense, low-jitter band, while the attacker-generated streams (brown and gray) exhibit dramatically higher jitter, large timing excursions, and inconsistent clock behavior. These malformed timing profiles reflect the attacker's inability to match the sender's RTP sequencing and timestamp cadence, resulting in significant packet-level disturbance without achieving decoder takeover. The visualization highlights the contrast between stable 2110 transport and the chaotic, out-of-profile packets produced by unsynchronized RTP injection.

RTP Stream Flip Scenario

During the payload-spoofing test, the receiver briefly flashed a green screen and then showed heavy macroblocking and visual glitches as the forged RTP packets started mixing into the stream. These artifacts lined up with a sharp spike in jitter and skew from the spoofed SSRC, which drifted so wildly that it reached mean jitter values above 32 ms and a maximum skew of more than 152 seconds, far outside the stable timing of the real sender. Even with these extreme timing errors, the receiver still accepted the attacker's packets because they matched

the legitimate multicast address and copied key RTP header fields like the SSRC, source port, and payload type. With both timing domains arriving at once, the decoder essentially fought to reconcile two competing streams, creating the flickering, green frames, and momentary corruption visible on screen. As the attack went on, the receiver occasionally recovered and snapped back to clean playout, reflecting its attempts to re-lock onto the genuine SSRC as the spoofed packets slowed. Meanwhile, the real sender's metrics stayed perfectly normal, zero loss, minimal jitter, stable drift, showing that all of the disruption was happening at the receiver due to RTP confusion, not because of a source or network issue. Overall, the experiment illustrates how forged RTP payloads can visibly distort video without tripping traditional transport-layer alarms, making this type of attack especially hard to detect.

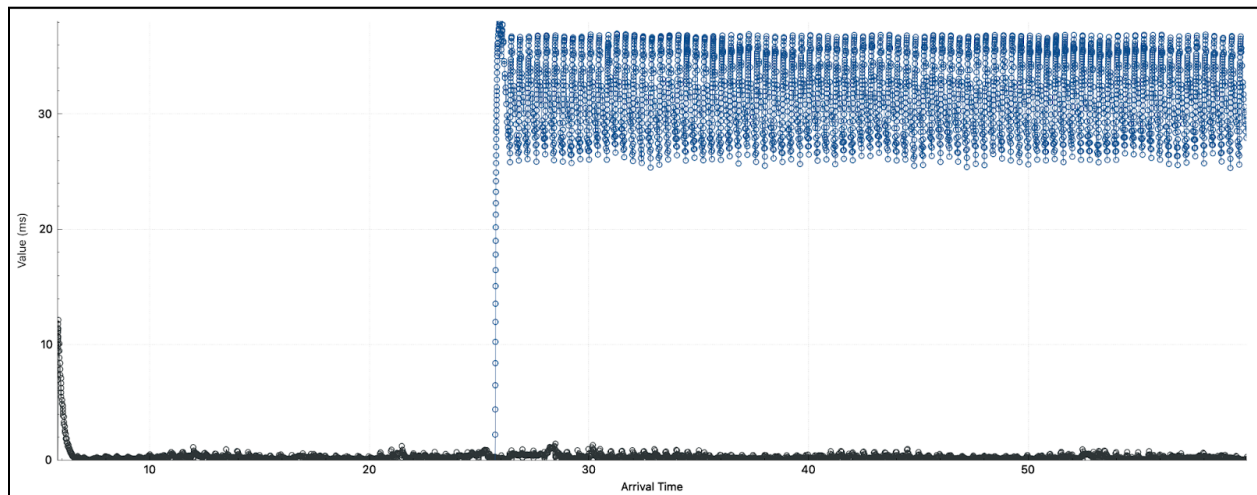


FIGURE 5. Sudden input of RTP Inter-Arrival Jitter Coinciding with Onset of Spoofed Packet Injection

Figure 5 shows a sharp rise in RTP jitter in a new stream at the exact moment the spoofed packets begin arriving. The baseline stream is nearly flat, but once the attacker injects forged RTP with conflicting timing, the new stream's jitter spikes into the 30–40 ms range and stays unstable. This timing disruption directly corresponds to the green-screen flashes and artifacting observed during the attack.



FIGURE 6. Video artifacting observed under RTP injection.

During the RTP injection test, the receiver briefly rendered corrupted frames as spoofed packets with mismatched timing overlapped the legitimate stream. The conflicting SSRC domains caused green-frame contamination and macroblocking, producing the characteristic artifacting shown in Figure 6.

NMOS Spoofing

This scenario demonstrated that a sustained HTTP flood against the mock NMOS registry produced measurable degradation to the control-plane without impacting media transport. Packet captures showed the attacker generating a continuous stream of HTTP GET requests, forcing the registry to serve hundreds of 404 responses per second and monopolizing its processing time. Despite the load, legitimate NMOS heartbeat POSTs from PI1, PI2, and PI3 continued to arrive, but their timing became irregular, with inter-arrival intervals stretching from the baseline 1-second cadence to multi-second gaps exceeding 9–15 seconds. This jitter indicates that the registry's request-handling loop was saturated, delaying heartbeat processing and reducing control-plane responsiveness. Importantly, none of the nodes fully disappeared from the registry, but the instability in heartbeat timing reflects a partial loss of liveness signaling. Interface counters and receiver PCAPs showed no packet loss, decode errors, or multicast disruption, confirming that the ST 2110 media-plane remained stable throughout the attack. This divergence between degraded control-plane behavior and unaffected media transport highlights a key operational risk: the NMOS registry can be impaired long before any visible impact appears on the media flows. Overall, the experiment shows that even a lightweight volumetric HTTP DoS is sufficient to destabilize NMOS-style orchestration services in a small-scale testbed.

Discussion

Across multiple adversarial scenarios, the most striking outcome was the extent to which ST 2110 media flows continued to exhibit stable visual quality even under conditions that heavily disrupted the control plane and transport-layer timing. This unexpected resilience suggests that the multicast dataplane, combined with receiver-side buffering and deterministic forwarding, can mask substantial network instability until attacks reach thresholds that directly interfere with packet ordering, timing coherence, or decoder lock. Taken together, the experiments reveal a clear separation between perceptible video degradation and underlying transport distortions, indicating that operators may be unaware of early-stage attacks unless sufficient telemetry, IGMP state monitoring, or RTP timing analysis is in place. This gap between observable playout quality and actual network health underscores the risk that broadcast engineers may overestimate system robustness, particularly in environments without PTP, redundant paths, or hardened control-plane services. Overall, the findings highlight that while 2110 media transport can appear visually stable under considerable stress, meaningful security hardening will require deeper visibility into timing behavior, proactive monitoring of control-plane anomalies, and safeguards that prevent attackers from exploiting this very resilience to remain undetected.

Potential Mitigation

Mitigation approaches for securing IP-based media systems have been widely discussed in industry and academic analyses, many of which emphasize the need to harden the traditionally unauthenticated components of ST 2110 and NMOS workflows. The European Broadcasting Union's security guidance stresses that strong network segmentation, strict access-control policies, and limitations on lateral movement are essential for preventing internal compromise in production environments [9]. These practices directly address the vulnerabilities demonstrated in the Layer-2 experiments, where ARP spoofing, MAC-table exhaustion, and VLAN boundary violations exploited trust-based Ethernet behavior. General network-security guidance from IEEE further reinforces the importance of authenticated address-resolution mechanisms, traffic filtering, and integrity protection to mitigate these link-layer attack vectors [12].

Complementing segmentation and L2 hardening, multicast-specific mitigations are critical for stabilizing ST 2110 distribution. Pecena's ST 2110 cybersecurity recommendations highlight practical defenses such as IGMP access controls, querier election hardening, and rate-limiting to prevent join/leave floods and rogue-querier manipulation [10]. Academic work by Sargent et al. demonstrates that IGMP remains inherently susceptible to forged membership reports and membership suppression attacks, supporting industry guidance for restricting IGMP messaging to authorized hosts and enforcing strict multicast registration policies [11]. These measures directly counter the multicast destabilization and forwarding-state inconsistencies observed in the IGMP attack scenarios.

At the application layer, both industry and academic sources emphasize the importance of securing NMOS APIs and ensuring RTP integrity. The NMOS security recommendations and broadcast-security analyses highlight that IS-04/IS-05 routing operations should be protected

with HTTPS, mutual authentication, and access-control policies to prevent unauthorized connection changes or resource manipulation [10]. While RTP itself lacks built-in security mechanisms, many studies recommend deploying network-level safeguards, such as ACLs, SSRC filtering, and per-stream whitelisting, to prevent packet injection, stream flipping, or sequence manipulation. Together, these mitigation strategies form a coherent defense-in-depth framework, addressing vulnerabilities from the link layer through the application layer and aligning directly with the failure modes demonstrated in this study.

#	Experiment Name	Primary Mitigations
1	Baseline 2110 flow	N/A (baseline reference only)
2	ARP spoof / gateway hijack	Dynamic ARP Inspection, static ARP bindings, MAC locking
3	MAC table flood	Port-security MAC limits, storm control, VLAN separation
4	Normal IGMP behavior	N/A (baseline reference only)
5	Rogue IGMP querier takeover	Fixed querier priority, IGMP ACLs, disable querier on hosts
6	Join/leave/report floods	IGMP rate-limiting, CoPP, restricted group membership
7	Basic RTP injection	Source ACLs, NMOS-controlled connections, SSRC/source filtering
8	Payload replacement	SRTP, sender whitelisting, continuity-counter monitoring
9	NMOS registry manipulation	HTTPS/TLS, API authentication, management-VLAN isolation

TABLE 2. Recommended Mitigations for Each Experiment Scenario

These mitigation techniques collectively illustrate how ST 2110 environments can be hardened against the vulnerabilities demonstrated in this study.

Limitations

This study, while effective in isolating and analyzing key vulnerabilities in ST 2110 and NMOS workflows, is limited by the scope and scale of the testbed, leaving several areas for deeper investigation in future work. The experiments were conducted on a simplified network topology with a small number of endpoints, which enabled tightly controlled observations but does not fully capture the complexity, traffic diversity, or operational scale of production broadcast environments. Additionally, the testbed did not incorporate PTP timing, redundant paths, or professional-grade switching hardware, meaning the results do not account for the resilience mechanisms, failure behaviors, or timing-domain interactions that typically influence real ST 2110 deployments. The absence of full NMOS ecosystem components, such as distributed registries, multi-controller orchestration, and large-scale resource graphs, also limits the ability to assess how control-plane attacks would propagate through more complex or federated deployments.

Beyond these structural constraints, the scope of the attack scenarios was limited to a subset of Layer-2, multicast, and application-layer behaviors, excluding several important threat classes that influence real-world ST 2110 resilience. Notably, the study did not examine PTP timing

attacks, 2022-7 path-redundancy manipulation, DHCP or DNS-based control-plane poisoning, or bandwidth-driven congestion scenarios that emerge when dozens of simultaneous high-bitrate flows share the same fabric. The experiments also focused on discrete sender-receiver pairs rather than large-scale, multi-flow environments in which contention, buffering behavior, and switch-queue dynamics may amplify or suppress adversarial effects. Finally, the analysis did not incorporate Quality-of-Experience metrics, such as latency, freeze duration, or perceptual degradation, leaving open questions about how different classes of attacks translate into operationally meaningful impact for live production workflows.

Future Work

Future work should extend this analysis into larger, more representative ST 2110 deployments that incorporate professional broadcast switches, PTP timing domains, and redundant architectures such as SMPTE ST 2022-7. Expanding the testbed to include multiple simultaneous video, audio, and ancillary streams would enable the study of congestion, queue contention, and timing interactions that cannot be observed in small-scale environments. Additional research should also evaluate adversarial behavior against full NMOS ecosystems, including distributed IS-04/IS-05 registries, multiple control applications, and cross-domain orchestration, to better understand how control-plane manipulation propagates in complex facilities. A dedicated line of investigation is also needed to assess PTP-layer attack surfaces, including delay attacks, rogue grandmasters, and boundary-clock manipulation. Finally, integrating perceptual Quality-of-Experience metrics and automated detection frameworks would help translate low-level anomalies into operator-relevant impact and inform the development of real-time monitoring and defense mechanisms for IP-based media workflows.

Conclusion

This work has shown that moving live media workflows from SDI to IP fundamentally reconfigures the security problem for broadcasters, exposing ST 2110 and NMOS systems to classes of attack that were largely irrelevant in closed baseband environments. Using a purpose-built 2110 testbed with segmented media, control, and management domains, and a structured progression of baseline, Layer-2, multicast/IGMP, RTP, and NMOS scenarios, the experiments provided an integrated, empirical view of how unauthenticated link-layer protocols, multicast control signaling, and API-driven orchestration behave under adversarial conditions. Rather than addressing each layer in isolation, the study traced how attacks at one layer propagate, or fail to propagate, into transport metrics, control-plane liveness, and ultimately visible playout.

Across all scenarios, a consistent pattern emerged: the IP media dataplane is often more resilient than its supporting control plane. ARP spoofing, rogue IGMP querier election, join/report floods, RTP injection, and NMOS HTTP abuse all produced measurable distortions in timing, routing logic, or registry behavior, yet the 2110-like stream frequently continued to appear “normal” at the receiver. In several cases, the multicast fabric maintained zero packet loss and stable SSRC continuity even while control-plane signaling degraded or switch CPU counters climbed sharply. This divergence between transport metrics and visual quality underscores a critical operational risk: engineers relying solely on subjective monitoring or

coarse loss/jitter thresholds may miss early-stage attacks that are already eroding safety margins and undermining orchestration reliability.

At the same time, the results confirm that many of the identified vulnerabilities are tractable with well-understood defenses. Techniques such as stronger Layer-2 hygiene (static or inspected ARP entries, MAC limits, storm control), hardened multicast behavior (fixed or authenticated querier roles, IGMP rate limiting, control-plane policing), and secure NMOS deployment (HTTPS/TLS, authentication, and strict scoping of registry access) directly target the failure modes observed in the threat lab. When combined with tighter segmentation between media, control, and management domains, these measures support a defense-in-depth posture that better aligns ST 2110 infrastructures with broader enterprise security architectures. Rather than treating media networks as a special case exempt from IT security practice, the experiments reinforce industry guidance that 2110 and NMOS should be protected using the same layered controls expected in other mission-critical IP systems.

Finally, the study highlights both the value and the limits of small-scale threat labs for guiding this transition. The miniature 2110 environment enabled controlled, repeatable attacks that would be impractical or unsafe in production facilities, and it captured nuanced behaviors, such as jitter-only impairments and registry timing skew, that are difficult to infer from static hardening checklists alone. At the same time, the absence of PTP, path redundancy, large-scale flow contention, and multi-controller NMOS ecosystems means that the results should be interpreted as lower-bound indicators of risk rather than exhaustive characterizations of all production topologies. Extending this work into larger fabrics, richer NMOS deployments, and PTP-aware timing domains will be essential to validate how these attack patterns evolve at scale and to refine monitoring, detection, and mitigation strategies. Nevertheless, the experiments presented here provide a concrete foundation: they demonstrate that IP-based media workflows can tolerate significant abuse before failing visibly, and they argue that secure operation in the 2110 era will depend as much on proactive telemetry and control-plane hardening as on the inherent robustness of the multicast dataplane itself.

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APPENDIX A: System Configuration

A.1 Testbed Overview

- Four Raspberry Pi endpoints (Pi1–Pi4) connected through a managed switch and Ubuntu router VM.
- Pi1: primary MPEG-TS over RTP sender.
- Pi2/Pi3: receivers for media-layer validation.
- Pi4: adversarial host for Layer-2, IGMP, and RTP attacks.
- Dell/Ubuntu workstation used for synchronized PCAP collection and log aggregation.

A.2 Network Topology

- Segmented into four VLANs:
 - *Media VLAN*: RTP, multicast flows.
 - *Control VLAN*: NMOS registry, signaling traffic.
 - *Management VLAN*: SSH, logging, system access.
- *Attacker Node* is bridged between *Media VLAN* and *Control VLAN*

A.3 Endpoint Configuration

- **Pi1 (Sender):**
 - FFmpeg pipeline streaming RTP to 239.x multicast.
 - Fixed SSRC, stable timestamp behavior.
Auto-started via systemd service.
 - Custom Scripting for ETH Interface snapshots
- **Pi2 (Receiver):**
 - VLC for playout, tshark for metrics.
Local logging enabled for jitter/loss capture.
 - Custom Scripting for ETH Interface snapshots
- **Pi3 (NMOS Node)**
 - tshark for metrics
 - Custom Scripting for:
 - ETH Interface snapshots
 - Heartbeat Journalctl Logging
- **Pi4 (Attacker):**
 - Python 3 + Scapy + custom scripts for:
 - ARP spoofing
 - MAC table flooding
 - IGMP join/leave manipulation
 - RTP injection and payload replacement

A.4 Router VM Configuration

- Ubuntu 22.04 LTS with four NICs mapped to VLANs and Internet
- Static addressing on each interface, NAT enabled for control-plane scenarios.
- Multicast routing disabled to ensure attacks remain Layer-2 within the Media VLAN.
- tcpdump used on all interfaces for synchronized captures.

A.5 Switch Configuration

- Managed L2 switch with the following profiles:
 - VLAN definitions for Media, Control, Management.
 - IGMP snooping active; querier disabled for baseline tests.
 - Port-to-VLAN bindings matching the testbed topology.
- Forwarding database (FDB) and multicast group membership tables monitored throughout experiments.

A.6 Synchronization & Logging

- All nodes synchronized to UTC using the multicap scheduling script.
- Capture windows aligned at the top of the minute for deterministic comparison across devices.
- Each capture event stored PCAPs, ARP tables, MFDB/FDB snapshots, and journalctl output in a structured directory.

APPENDIX B: Procedures & SOPs

B.1 Packet Capture Procedure

- All captures scheduled using the unified multicap script.
- Start time aligned to the next UTC minute.
- Each node launched a tcpdump instance on its primary interface.
- Router VM captured **all interfaces** for complete visibility.
- Capture duration standardized (typically 60 seconds per run).
- Output stored in run-specific folders with timestamps and device labels.

B.2 Sender Startup Procedure

- Pi1 sender operated as a systemd service.
- Service auto-started on boot; manually restartable for experiments.
- RTP pipeline fixed to multicast group + port with stable SSRC.
- Logging captured service status, restarts, and timestamp alignment.

B.3 Baseline Measurement Procedure

- Verify connectivity on all VLANs.
- Confirm IGMP membership for receiver ports.

- Validate sender stability (jitter, loss = 0).
- Take initial PCAPs and store as baseline references.

B.4 ARP Spoofing Procedure

- Pi4 launched ARP response injector targeting router and endpoints.
- Spoofed “is-at” packets sent repeatedly to overwrite ARP caches.
- Logs captured ARP tables before, during, after attack.
- Observed NMOS heartbeat loss and routing interruption.

B.5 MAC Table Flood Procedure

- Pi4 transmitted large volumes of randomized-source Ethernet frames.
- Switch FDB monitored for churn and overflow behavior.
- Media VLAN observed for jitter/loss impacts and traffic flooding.
- Capture recorded switch and endpoint responses.

B.6 IGMP Manipulation Procedure

- Pi4 generated IGMPv2/v3 joins, leaves, and rogue queries.
- MFDB tables captured before and after IGMP bursts.
- Receivers monitored for join instability or group changes.
- Metrics recorded changes in packet loss and jitter.

B.7 RTP Injection Procedure

- Pi4 injected spoofed RTP packets matching sender’s SSRC or port.
- Payload alternated between noise and crafted frames.
- Wireshark RTP analysis recorded timing anomalies (jitter/skew).
- Receiver output documented visible artifacting or corruption.

B.8 Payload Replacement Procedure

- Pi4 replaced legitimate payload with alternate encoded frames.
- Both legitimate and spoofed SSRCs appeared on-wire simultaneously.
- Receiver evaluated for dominance behavior and decode artifacts.
- Stability measured post-injection to detect recovery behavior.

B.9 NMOS Spoofing Procedure

- Pi4 crafted fake NMOS IS-04/IS-05 style JSON registrations to the registry.
- Queried registry state to confirm unauthorized node insertion.
- Checked whether receivers attempted to subscribe to false flows.
- Monitored for control-plane confusion or routing errors.

APPENDIX C: Scripts & Tools

C.1 Capture Scripts

- **Multicap synchronization script:**
 - Schedules PCAP captures to start at the next UTC minute.
 - Triggers tcpdump on all Pis and router VM simultaneously.
 - Logs run ID, duration, host list, and configuration parameters.
 - Creates structured output folders for each experiment run.
- **Router all-interface capture:**
 - tcpdump started on every NIC to record all VLANs.
 - Ensures cross-VLAN visibility during Layer-2 attacks.
 - Useful for validating FDB/MFDB state during floods or spoofing.

C.2 Attack Scripts

- **ARP Spoofing Script (Python/Scapy):**
 - Sends continuous forged ARP replies with attacker MAC.
 - Targets router IP and endpoint ARP caches.
 - Includes options for unicast or broadcast poisoning.
- **MAC Flood Script:**
 - Generates high-rate frames with randomized source MACs.
 - Causes CAM/FDB table exhaustion on the switch.
 - Adjustable rate and payload size.
- **IGMP Manipulation Script:**
 - Sends IGMP queries, joins, and leaves artificially.
 - Can simulate rogue querier takeover.
 - Supports both IGMPv2/v3 message formats.
- **RTP Injection Script:**
 - Crafts RTP packets matching legitimate SSRC or ports.
 - Adjustable payload, timestamps, and sequence numbers.
 - Used for SSRC collision, jitter manipulation, and injection tests.
- **Payload Replacement Script:**
 - Replaces media payload with alternate video frames.
 - Spoofs SSRC and timing to collide with real sender.
 - Generates visible corruption and green-frame artifacts.

C.3 Utility Tools

- **ARP/MFDB/FDB Snapshot Tools:**
 - Captures live ARP tables on Pis.
 - Queries switch MFDB and FDB tables for group membership state.
 - Logged before, during, and after attack scenarios.
- **NMOS JSON Utilities:**
 - Simple curl-based scripts for IS-04/IS-05 POST/DELETE operations.
 - Used to inject fake nodes, senders, and flows into the registry.
 - Includes heartbeat simulators for spoofed devices.

- **Log Aggregation Scripts:**
 - Collect journalctl, dmesg, tcpdump summaries, and system timestamps.
 - Bundles experiment metadata into run folders for reproducibility.

C.4 Supporting Tools (External)

- *tshark / Wireshark*: RTP analysis, jitter/skew metrics, IGMP inspection.
- *ffmpeg*: Media generation, payload crafting, packet timestamp manipulation.
- *VLC*: Receiver reference for visual artifact validation.

APPENDIX D: Logs & PCAP Artifacts

D.1 Representative PCAP Summaries

- **Baseline (Experiment 1):**
 - Zero packet loss, stable jitter (<0.3 ms), no SSRC conflicts.
 - IGMP membership is consistent and unchanged throughout capture.
- **ARP Spoofing (Experiment 2):**
 - Sudden ARP cache overwrite events visible in PCAP.
 - ICMP failures and NMOS heartbeat drops aligned with the poisoning window.
 - RTP freeze and packet-gap clusters observed at attack onset.
- **MAC Table Flood (Experiment 3):**
 - High volume of randomized MAC source frames.
 - Switch flooding behavior captured: multicast replication to unintended ports.
 - No SSRC changes; jitter spikes during flooding intervals.
- **IGMP Manipulation (Experiments 4–7):**
 - Rapid IGMP join/leave bursts and rogue queries recorded.
 - MFDB instability producing short-lived multicast path changes.
 - Occasional micro-freezes in RTP continuity counter.
- **RTP Injection & Payload Spoofing (Experiments 8–9):**
 - Multiple SSRCs visible on-wire (legitimate + attacker).
 - Abnormal jitter values and negative skew on spoofed packets.
 - Payload corruption segments corresponding to visible artifacting.
- **DoS Flooding (Experiments 10–11):**
 - UDP flood saturating VLAN bandwidth.
 - Receiver jitter rise >10 ms and packet loss bursts.
 - NMOS control-plane delays and timeout events captured.
- **NMOS Spoofing (Experiment 12):**
 - Unauthorized POST requests to registry captured in PCAP.
 - Fake node, sender, and flow announcements over HTTP.
 - Receiver attempts to query non-existent senders.

D.2 Key Log Excerpts

- **journalctl:**

- Interface flaps, ARP updates, multicast join requests, and systemd sender restarts.
- **dmesg:**
 - Kernel-level warnings during heavy packet rates (MAC flood / DoS).
 - NIC statistics showing receive overruns.
- **ARP Table Snapshots:**
 - Correct mappings (baseline) vs spoofed “dc:a6:...” attacker MAC (ARP test).
- **MFDB / FDB Snapshots:**
 - Membership list changes during IGMP churn.
 - Excessive FDB entries during CAM exhaustion.
- **NMOS Registry Logs:**
 - Heartbeat intervals.
 - Missing-node timeouts during Layer-2 outages.
 - Unauthorized IS-04/IS-05 registrations during spoofing tests.

D.3 Metrics Tables

- **Per-experiment RTP statistics:**
 - Max jitter, mean jitter, max delta, skew, clock drift.
- **Sender vs Receiver Comparison:**
 - Errors isolated to receiver timing domain during spoofing attempts.
 - Sender metrics remain stable across all adversarial scenarios.
- **Packet Loss Timelines:**
 - Burst loss aligned with ARP poisoning.
 - Sporadic loss during IGMP flooding.
 - Sustained loss in DoS media-flood tests.
- **SSRC Behavior:**
 - No collisions baseline.
 - Dual-SSRC presence during spoofing experiments.
- **Control-plane Metrics:**
 - NMOS heartbeat intervals.
 - Registration delays under load.
 - Query/response latency spikes during HTTP flood.

D.4 Visual Artifacts & Event Markers

- **Green-frame contamination:**
 - Documented in payload-spoofing test.
 - Timestamped alongside RTP jitter spikes.
- **Freeze-frames / continuity jumps:**
 - Occur during ARP gateway hijack and IGMP instability.
- **Macroblocking bursts:**
 - Visible when spoofed timestamps collide with legitimate packets.

APPENDIX E: Supplemental Repository

- All scripts, PCAP files, log archives, switch snapshots, Wireshark graphs, and experiment datasets are available in the project's public repository.
- The repository contains:
 - Full-resolution figures and diagrams
 - Complete attack/defense scripts
 - Sender/receiver pipelines
 - Multicap capture outputs
 - VLAN maps and switch configurations
 - Raw and processed metrics for all experiments

GitHub Repository:

https://github.com/QuarksAndTings/TechnicalPaper_2110CyberSecurity_Experiment

This repository serves as the complete supplemental dataset for reproducibility and extended analysis.